

Metal Film Resistors

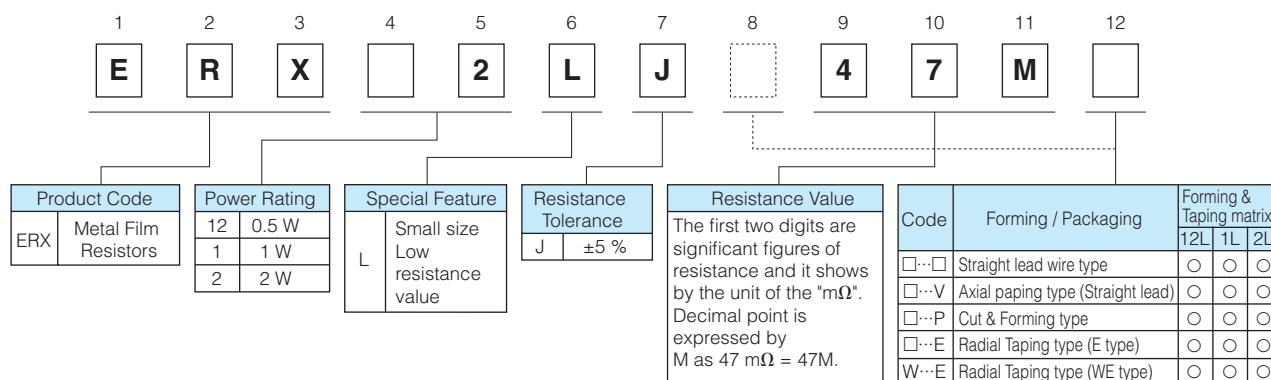
Type: **ERXL (Low Resistance Value)**
(0.5 W, 1 W, 2 W)



Features

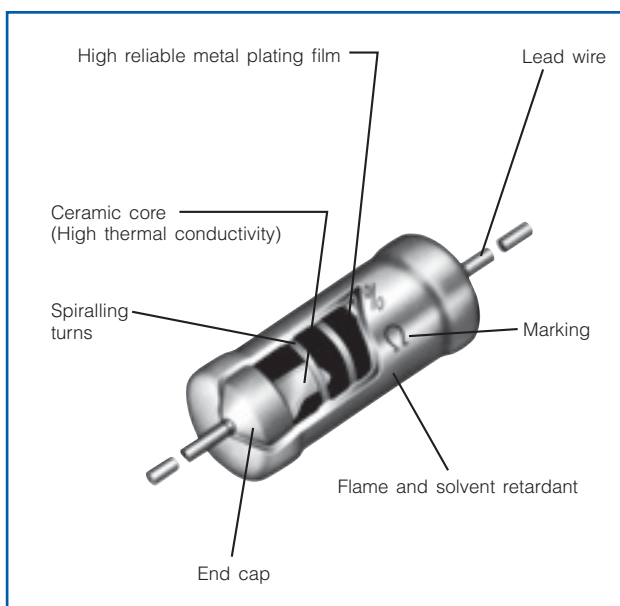
- Miniaturized
- Non-flammable
- Automatic Insertion
- RoHS compliant

Explanation of Part Numbers

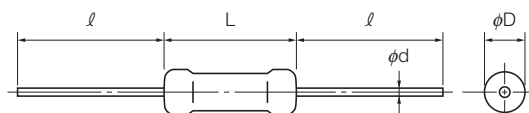


The above example shows a small size and low resistance value metal film resistor, 2 W power rating, resistance value of 47 m ohms, tolerance $\pm 5\%$, and package of standard bulk packing.

Construction



Dimensions in mm (not to scale)



Part No.	Dimensions (mm)				Mass (Weight) [g/pc.]
	L	ϕD	ℓ	ϕd	
ERX12L	$6.35^{+0.65}_{-0.35}$	$2.3^{+0.5}_{-0.3}$	$30.0^{\pm 3.0}$	$0.65^{\pm 0.05}$	0.26
ERX1L	$9.00^{+1.50}_{-1.00}$	$2.8^{\pm 0.5}$	$30.0^{\pm 3.0}$	$0.65^{\pm 0.05}$	0.33
ERX2L	$12.00^{+1.50}_{-1.00}$	$4.0^{\pm 1.0}$	$30.0^{\pm 3.0}$	$0.80^{\pm 0.05}$	0.66

Design and specifications are each subject to change without notice. Ask factory for the current technical specifications before purchase and/or use. Should a safety concern arise regarding this product, please be sure to contact us immediately.

02 Sep. 2014

Panasonic Metal Film Resistors, Low Resistance Value

Ratings

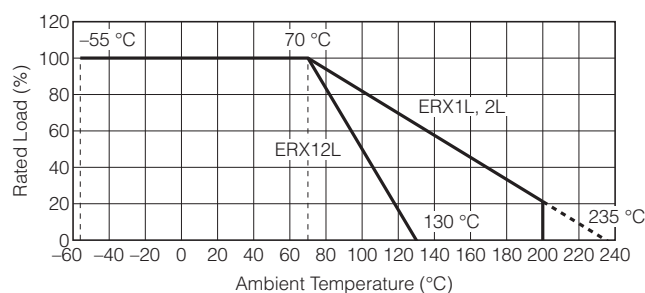
Part No.	Power Rating at 70 °C ⁽¹⁾ (W)	Dielectric Withstanding Voltage (VAC)	Res. Tol. (%) ⁽²⁾	Resistance Range (Ω) ⁽²⁾		T.C.R. (×10 ⁻⁶ /°C)	Standard Resistance Value
				min.	max.		
ERX12L	0.5	350	J (±5)	22 m	82 m	22 to 39 mΩ=±1000 47 to 82 mΩ=± 500	E12
ERX1L	1	350	J (±5)	22 m	82 m		E12
ERX2L	2	600	J (±5)	22 m	82 m		E12

(1) Rated Continuous Working Voltage (RCWV) shall be determined from $RCWV = \sqrt{\text{Power Rating} \times \text{Resistance Value}}$.

(2) Resistance tolerance and resistance range is of use besides range listed, please inquire.

Power Derating Curve

For resistors operated in ambient temperatures above 70 °C, power rating shall be derated in accordance with the figure on the right.



■ As for Packaging Methods and / or cut formed leads,

Please see Metal (Oxide) Film Resistors Packaging Methods

⚠ Safety Precautions

The following are precautions for individual products. Please also refer to the common precautions for Fixed Resistors in this catalog.

1. Transient voltage

If there is a possibility that the transient phenomenon (significantly high voltage applied in a short time) may occur or that a high voltage pulse may be applied, make sure to evaluate and check the characteristics of Metal Film Resistors (hereafter called the resistors) mounted on your product rather than only depending on the calculated power limit or steady-state conditions to complete the design or decide to use the resistors.

- The resistors are covered with a special coating. Do not apply shock or vibration to them, or pinch them with long-nose pliers. Otherwise, the resistors may be damaged.
- Do not apply excessive tension to the lead-connected sections. When bending the lead wire, do not apply excessive stress to the resistors and provide the wire with a natural curvature.
- Do not brush the resistors during or after the cleaning process, which may be conducted after soldering. Otherwise, the coating film may be damaged.